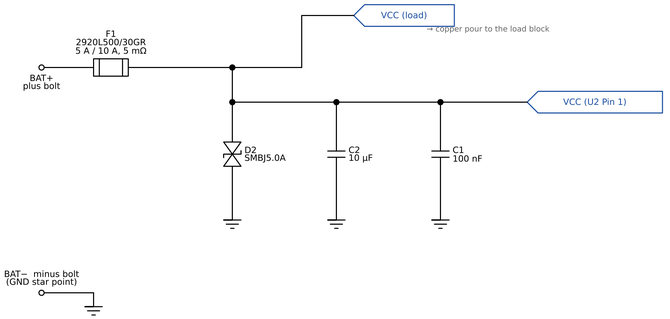
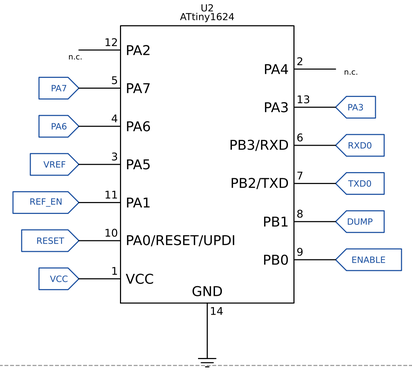
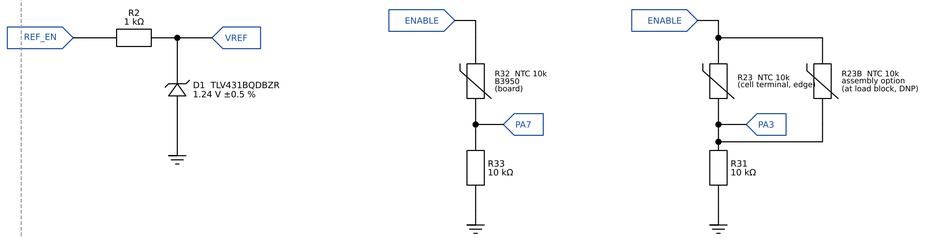
1 Cell connection, fuse, MCU supply (star point at F1)



2 Microcontroller ATtiny1624-SSU (U2)

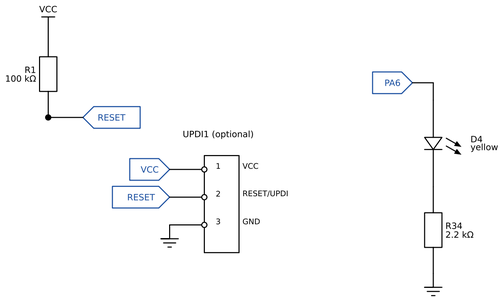


3 Voltage reference and temperature sensors

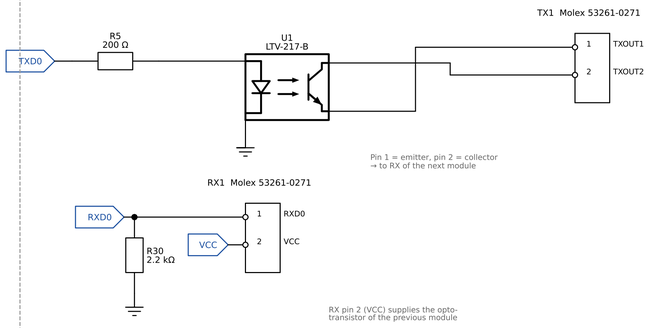


R23 sits at the bottom edge right next to the plus bolt. Underneath (bottom side) a copper area on the plus terminal acts as a thermal bridge → R23 measures the terminal / cell temperature. R23B = alternative position at the load block (not fitted). Fit R23 OR R23B.

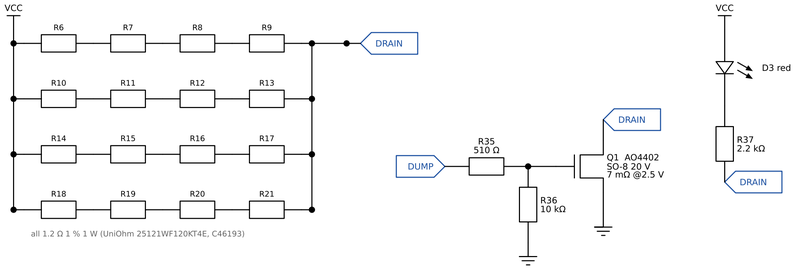
4 Reset, programming (UPDI), status LED



5 Communication (isolated), Molex PicoBlade 1.25 mm connectors



6 Balancing load 16 × 1.2 Ω 2512 (4S4P = 1.2 Ω, approx. 3 A @ 3.65 V)



all 1.2 Ω 1 % 1 W (UniOhm 25121WF120KT4E, C46193)

Net names (blue) connect points of the same name. Bottom GND plane connects to the load current path only at the minus bolt.